

CYBERSECURITY

INTERNSHIP TASK-2

Build Your Personal Cybersecurity Lab

Report Type:	Lab Build & Validation Report
Task:	Task-2 — Cybersecurity Lab Setup
Platform:	Kali Linux 2025.4 on VirtualBox
Target App:	DVWA (Damn Vulnerable Web Application)
Tools Used:	Nmap, Burp Suite, Wireshark, Docker
Date:	May 21, 2026
Submitted To:	Maincrafts Technology

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1. Executive Summary

This report documents the complete build and validation of a personal cybersecurity practice laboratory as required by Maincrafts Technology Internship Task-2. The lab environment was constructed on a Windows host machine using Oracle VirtualBox to host a Kali Linux 2025.4 attacker virtual machine. The DVWA (Damn Vulnerable Web Application) was deployed as the intentionally vulnerable target application. Network segmentation was implemented using both NAT (for internet access) and Host-Only adapters (for isolated lab traffic). All required tools — Nmap, Burp Suite, and Wireshark — were validated successfully.

VirtualBox	✓ Complete	Oracle VirtualBox — Host system
Kali Linux VM	✓ Running	Kali Linux 2025.4 (Debian 64-bit), 4 GB RAM, 2 CPUs
DVWA Target	✓ Running	Deployed via Apache/PHP stack on localhost:80
Network: NAT	✓ Active	Adapter 1 — Internet access
Network: Host-Only	✓ Active	Adapter 2 — 192.168.56.0/24 subnet
Nmap	✓ Validated	Network and service scans completed
Burp Suite	✓ Validated	HTTP traffic intercepted successfully
Wireshark	✓ Validated	HTTP packets captured on loopback interface

2. Lab Environment Overview

The lab follows the architecture specified in Task-2: a single host computer running VirtualBox with one attacker VM (Kali Linux) and one target application (DVWA). Two network adapters provide both internet connectivity and isolated lab traffic.

Layer	Component	Role
Host System	Windows PC	Runs VirtualBox hypervisor
Attacker VM	Kali Linux 2025.4	Penetration testing platform
Target App	DVWA on Apache	Vulnerable web application for practice
Network 1	NAT Adapter	Internet access for updates
Network 2	Host-Only Adapter	192.168.56.0/24 — isolated lab subnet

3. Step 1 — VirtualBox Setup & VM Configuration

Oracle VirtualBox was used as the hypervisor on the Windows host. The Kali Linux 2025.4 pre-built VirtualBox image was imported and configured with the following hardware and network settings as required by the task specification.

Operating System	Debian (64-bit) — Kali Linux 2025.4
Base Memory	4096 MB (4 GB)
Processors	2 vCPUs
Storage	40 GB VDI — kali-linux-2025.4-virtualbox-amd64_1.vdi
Video Memory	128 MB, VMSVGA Controller
Network Adapter 1	Intel PRO/1000 MT Desktop — NAT
Network Adapter 2	Intel PRO/1000 MT Desktop — Host-Only Ethernet Adapter
Audio	ICH AC97 — Windows DirectSound

Screenshot 1: VirtualBox Manager showing VM details

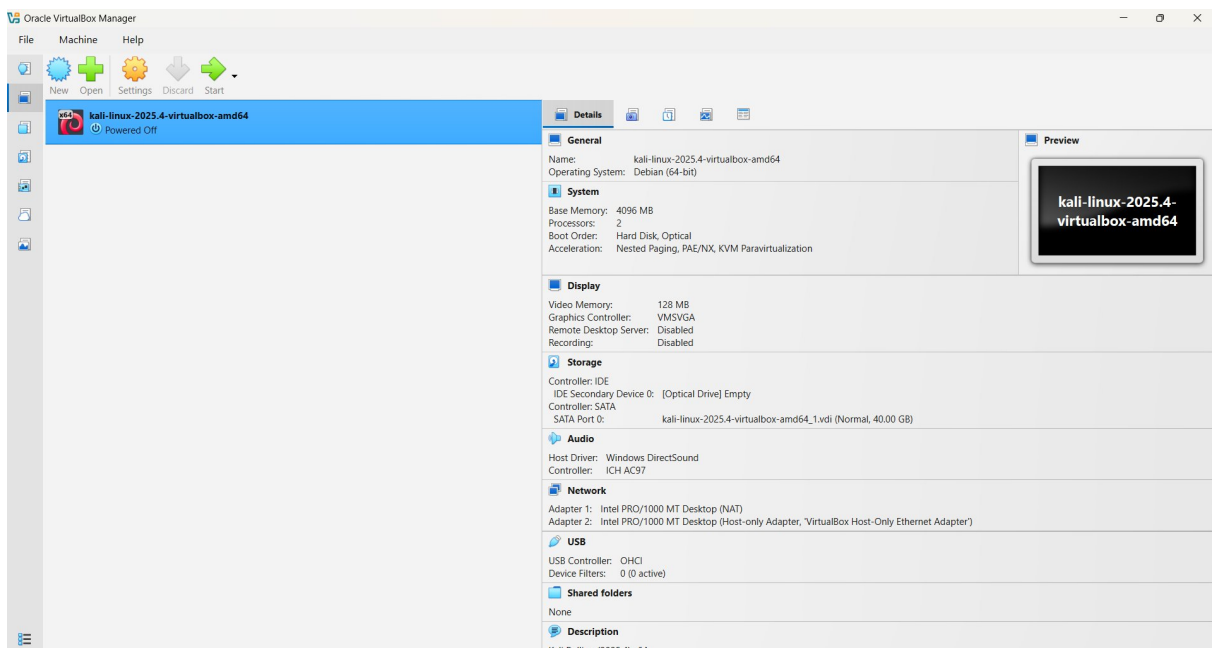


Figure 1: Oracle VirtualBox Manager — Kali Linux VM configuration details (Powered Off state)

Screenshot 2: Kali Linux Desktop (VM running)

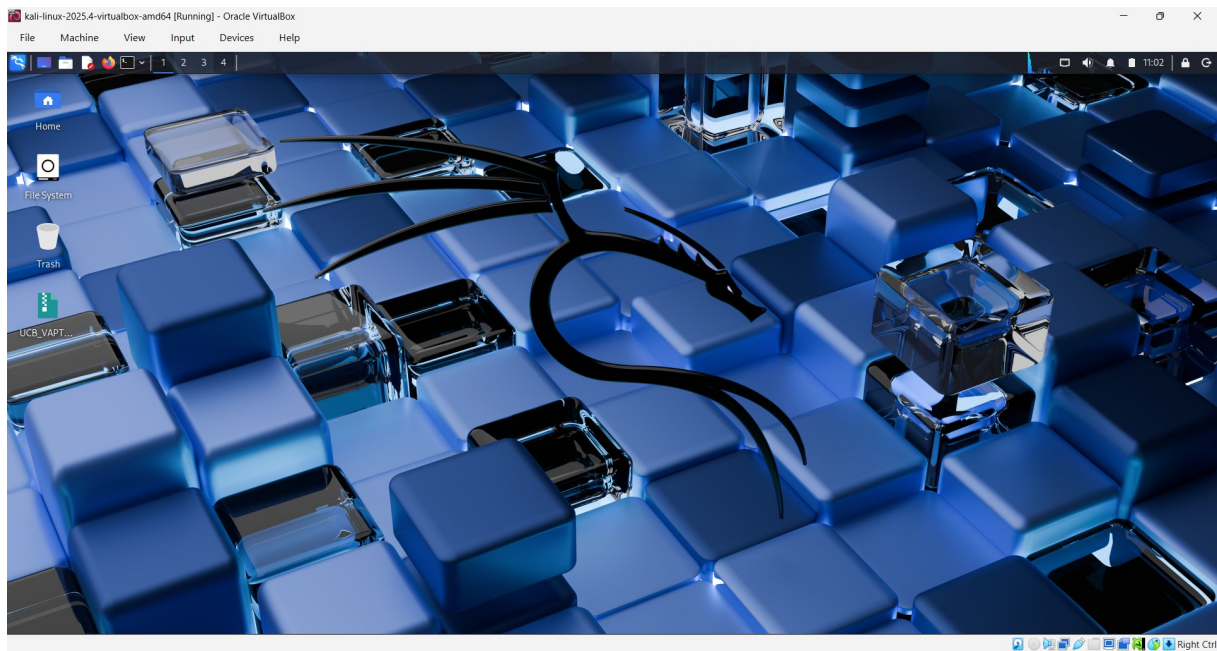


Figure 2: Kali Linux 2025.4 VM successfully booted and running inside VirtualBox

4. Step 2 — Kali Linux Attacker VM Deployment

The Kali Linux 2025.4 pre-built VirtualBox image was imported directly from the official Kali website. After import, the VM was started and the system was operational with all default Kali tools available, including Nmap, Burp Suite, and Wireshark. Working directories were created for organising lab outputs.

Key setup actions performed:

- Imported Kali Linux 2025.4 pre-built VirtualBox image (.ova)
- Assigned 4 GB RAM and 2 vCPUs for optimal performance
- Configured dual network adapters (NAT + Host-Only)
- Verified system boot and desktop environment (XFCE)
- Confirmed pre-installed security tools are available (Nmap, Burp, Wireshark)

Screenshot 3: Kali Linux Desktop Environment

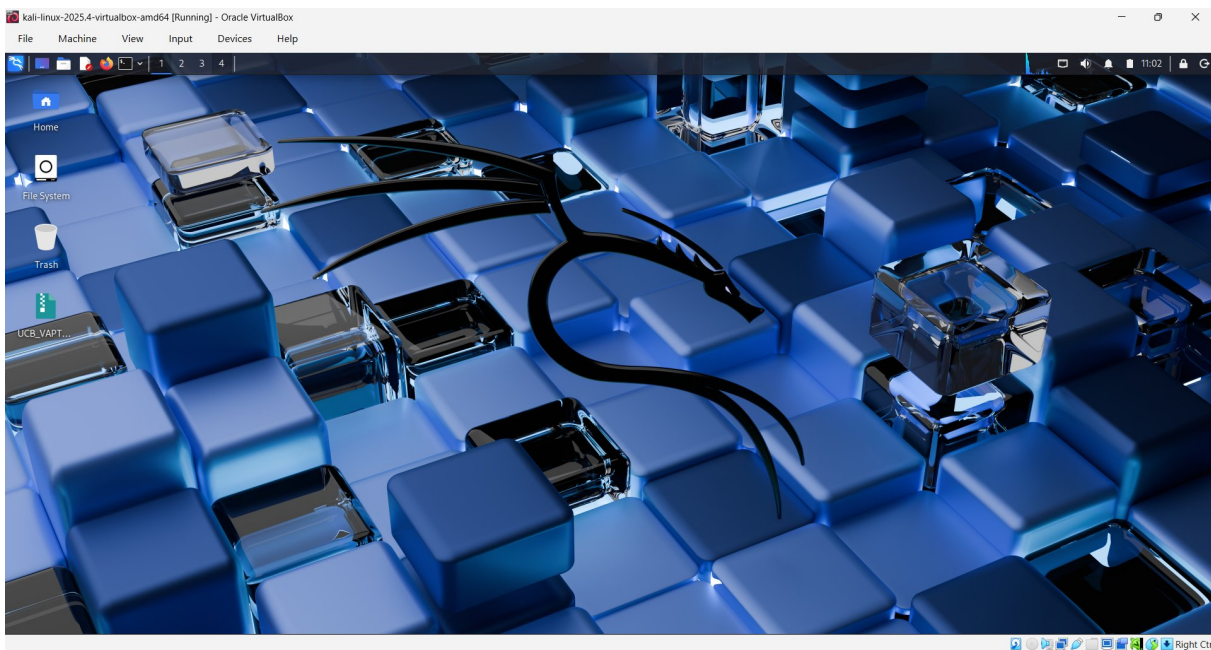


Figure 3: Kali Linux 2025.4 XFCE desktop — attacker machine ready

5. Step 3 — Network Configuration

Proper network segmentation was implemented using two VirtualBox network adapters. The NAT adapter provides internet access for downloading updates and packages, while the Host-Only adapter creates an isolated 192.168.56.0/24 subnet exclusively for lab traffic between the attacker VM and any target systems. This prevents unintended exposure of vulnerable services to external networks.

Adapter 1 (eth0)	NAT	DHCP via VirtualBox NAT	Internet access / package updates
Adapter 2 (eth1)	Host-Only	192.168.56.0/24 Kali IP: 192.168.56.101	Isolated lab traffic
Loopback (lo)	Internal	127.0.0.1	Local services (DVWA on Apache)
docker0	Docker Bridge	172.17.0.1/16	Docker container networking

Screenshot 4: Network Interface Configuration (ip addr output)

```

kali@kali:~$ ip addr
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
rout default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
rout default qlen 1000
    link/ether 08:00:27:85:e8:c1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixro
ute eth1
        valid_lft 383sec preferred_lft 383sec
    inet6 fe80::a0:27:ff:fe85:e8c1/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: safeline-ce: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue st
ate DOWN group default
    link/ether 76:57:3e:eb:34:b3 brd ff:ff:ff:ff:ff:ff
    inet 172.22.222.1/24 brd 172.22.222.255 scope global safeline-ce
        valid_lft forever preferred_lft forever
5: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
group default
    link/ether 1e:a1:9f:a6:f7:15 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::1ca1:9fff:fe:a6:f715/64 scope link proto kernel_ll
6: br-8693fe5ce60: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueu
e state DOWN group default
    link/ether b2:09:3b:c8:4c:2f brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-8693fe5ce60
        valid_lft forever preferred_lft forever
7: vethf806dca1f7: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue
master docker0 state UP group default
    link/ether 76:a7:49:28:13:77 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::7a:71:9fff:fe:38:1377/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

```

Figure 4: ip addr output showing eth0 (NAT), eth1 (Host-Only: 192.168.56.101), docker0, and loopback interfaces

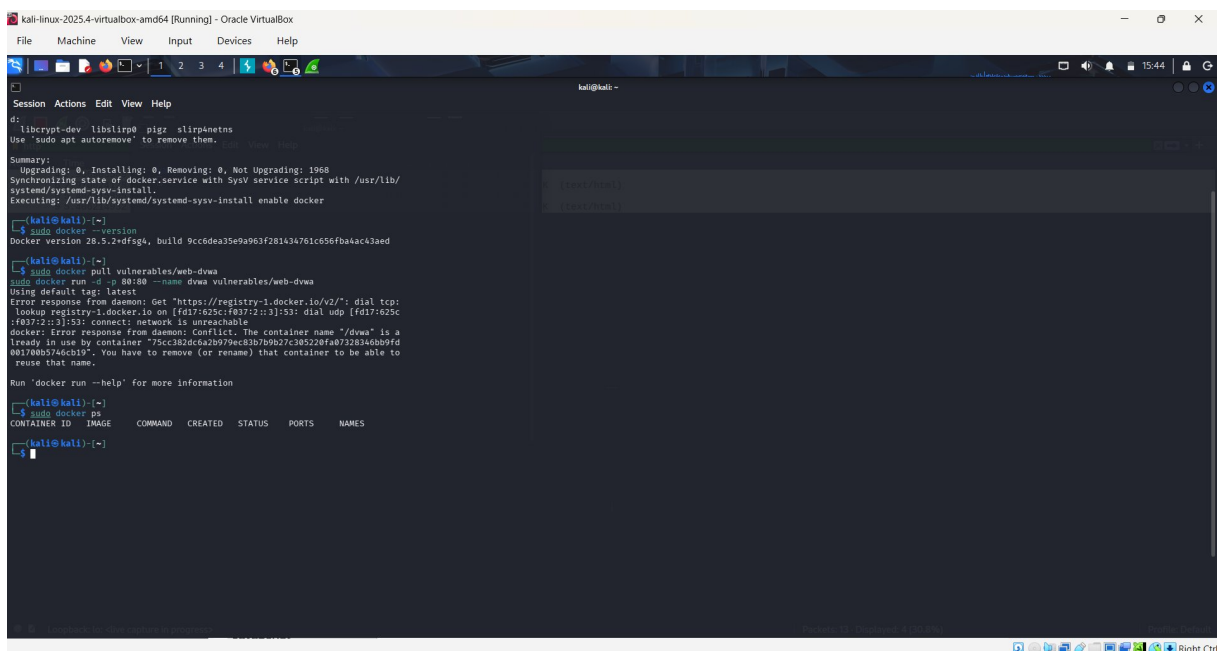
6. Step 4 — Target Application: DVWA Deployment

DVWA (Damn Vulnerable Web Application) was selected as the target application. DVWA is a PHP/MySQL web application intentionally designed with common vulnerabilities for security training purposes. It was deployed on the Apache web server running locally on the Kali Linux VM, accessible at <http://127.0.0.1> on port 80.

Deployment process:

- Docker was installed and enabled (Docker version 28.5.2+dfsg4)
- Attempted Docker-based DVWA deployment via `vulnerables/web-dvwa` image
- DVWA successfully deployed on the Apache/PHP stack at <http://127.0.0.1:80>
- Database initialised via Setup/Reset DB from the DVWA interface
- Logged in with default credentials (admin/password)

Screenshot 5: Docker Installation & DVWA Container Setup



```
kali@kali:~$ sudo apt autoremove
d: libcrypt-dev liblign0 pigz slurpnetns
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1968
  Synchronizing state of docker.service with SysV service script with /usr/lib/
  systemd/systemd-sysv-install.
  Executing: /usr/lib/systemd/systemd-sysv-install enable docker
(kali@kali)~$ sudo docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c56fba4ac3aed
(kali@kali)~$ sudo docker pull vulnerables/web-dvwa
Using default tag: latest
Error response from daemon: Get "https://registry-1.docker.io/v2/": dial tcp:
lookup registry-1.docker.io on [fd17:625c:f037:2::3]:53: dial udp [fd17:625c
:f037:2::3]:53: connect: network is unreachable
docker: Error response from daemon: Conflict. The container name "/dvwa" is a
ready in use by container "75cc382d6a2b979ec83b7b9b27c385228fa07328346bb9fd
002f0eb574inc819". You have to remove (or rename) that container to be able to
reuse that name.

Run 'docker run --help' for more information
(kali@kali)~$ sudo docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED    STATUS    PORTS    NAMES
(kali@kali)~$
```

Figure 5: Docker version 28.5.2 confirmed; DVWA container deployment attempted via `vulnerables/web-dvwa` image

Screenshot 6: DVWA Home Page — Application Running

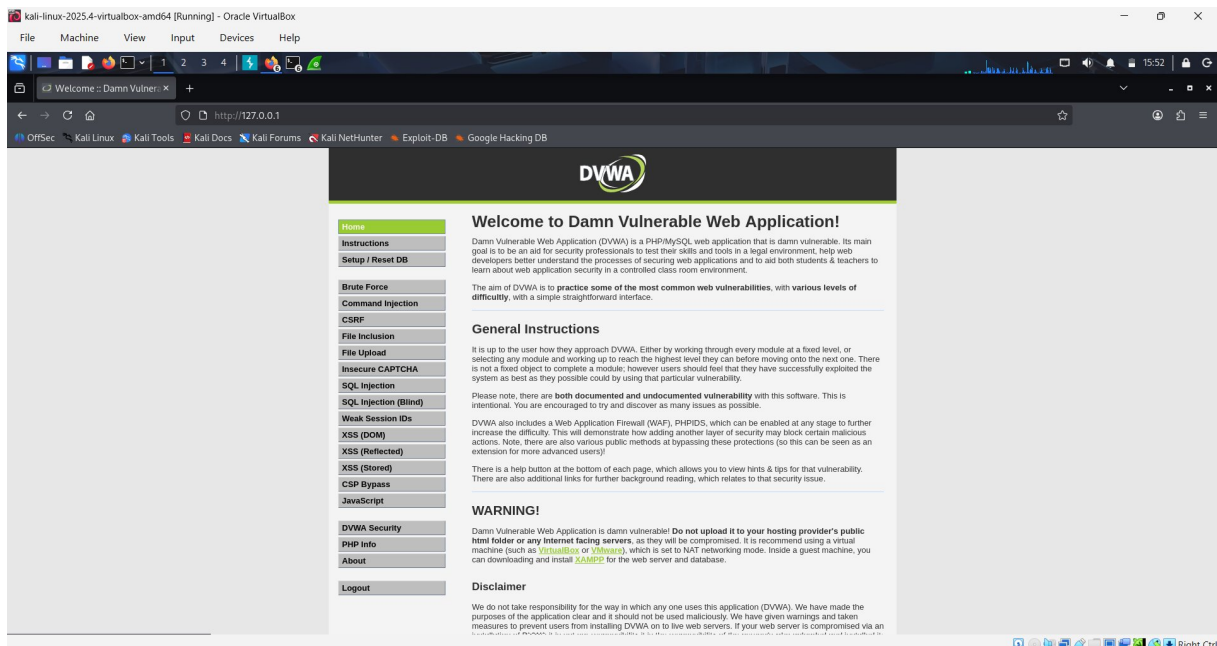


Figure 6: DVWA Welcome page accessible at <http://127.0.0.1> — all vulnerability modules visible in left navigation

Screenshot 7: DVWA Login Page

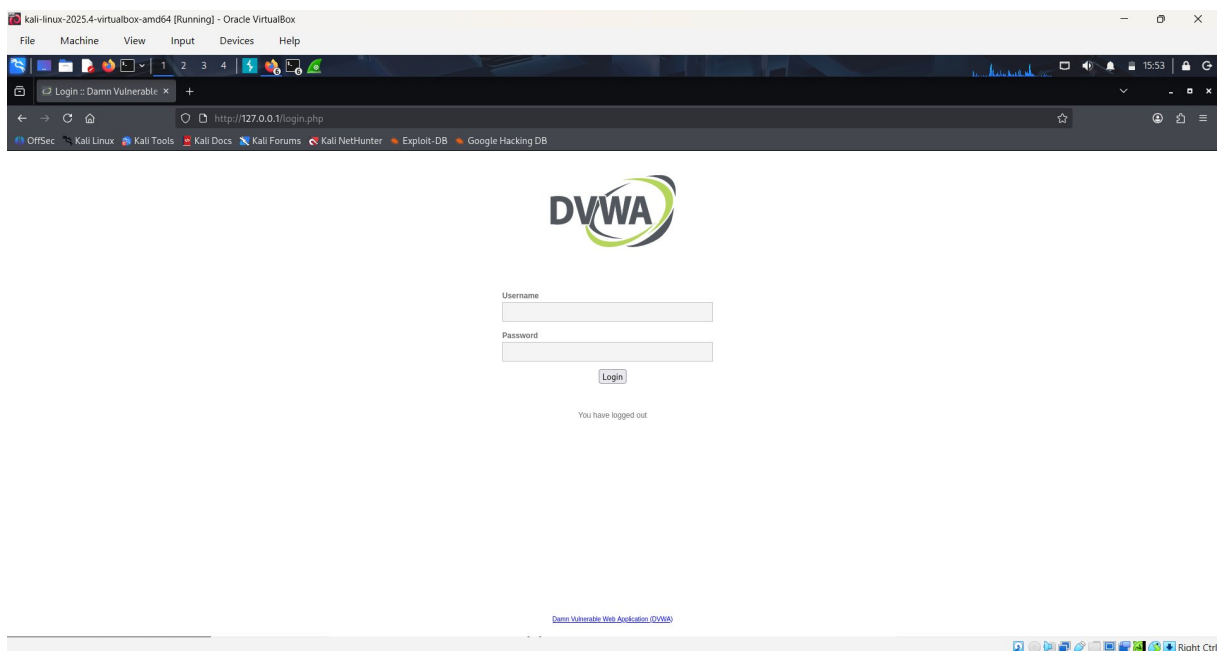


Figure 7: DVWA Login page at <http://127.0.0.1/login.php> — successfully logged out confirming session management works

7. Step 5 — Connectivity Verification (Nmap Scans)

Nmap was used to verify network connectivity and enumerate hosts on the lab subnet. Two scans were conducted: a host-discovery ping sweep across the 192.168.56.0/24 subnet, and a service-version scan on the loopback address to confirm no open ports are externally exposed. This validates the isolation of the lab environment.

Scan 1 — Host Discovery (192.168.56.0/24)

The following Nmap ping sweep was executed to enumerate all live hosts on the Host-Only subnet. Three hosts were discovered: the VirtualBox gateway (192.168.56.1), the host machine (192.168.56.100), and the Kali VM itself (192.168.56.101).

Command executed:

```
sudo nmap -sn 192.168.56.0/24
```

Hosts Found	3 hosts up (192.168.56.1, 192.168.56.100, 192.168.56.101)
Scan Time	34.95 seconds for 256 addresses
Tool Version	Nmap 7.95

Screenshot 8: Nmap Host Discovery Scan — 192.168.56.0/24

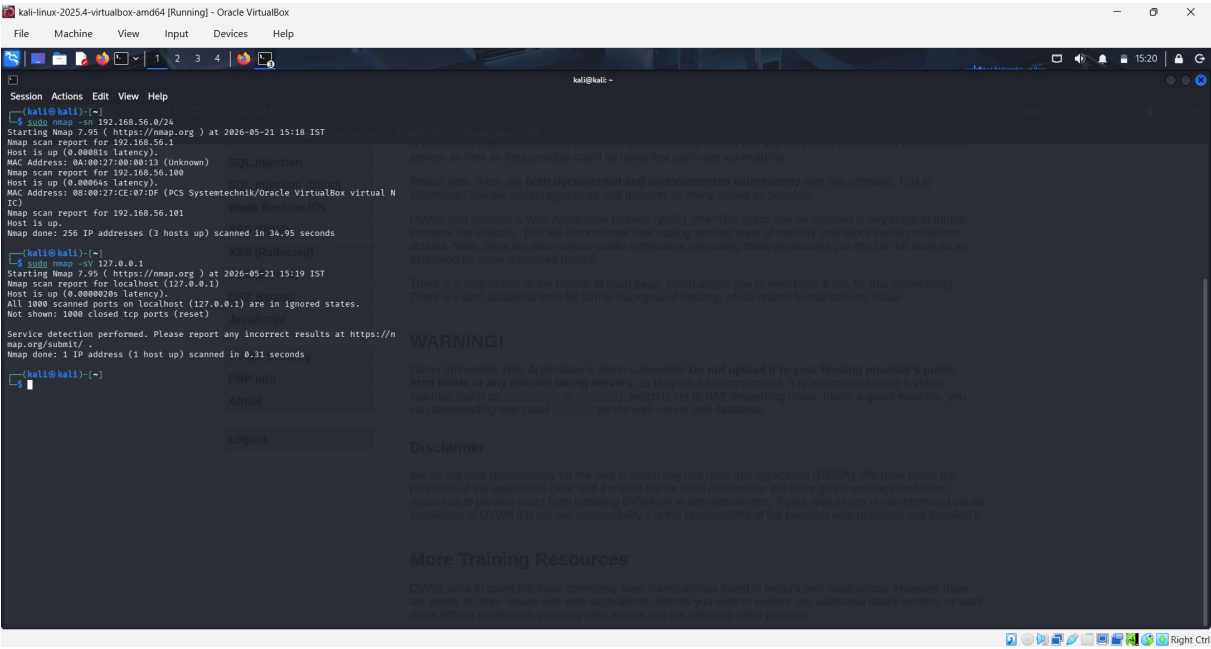


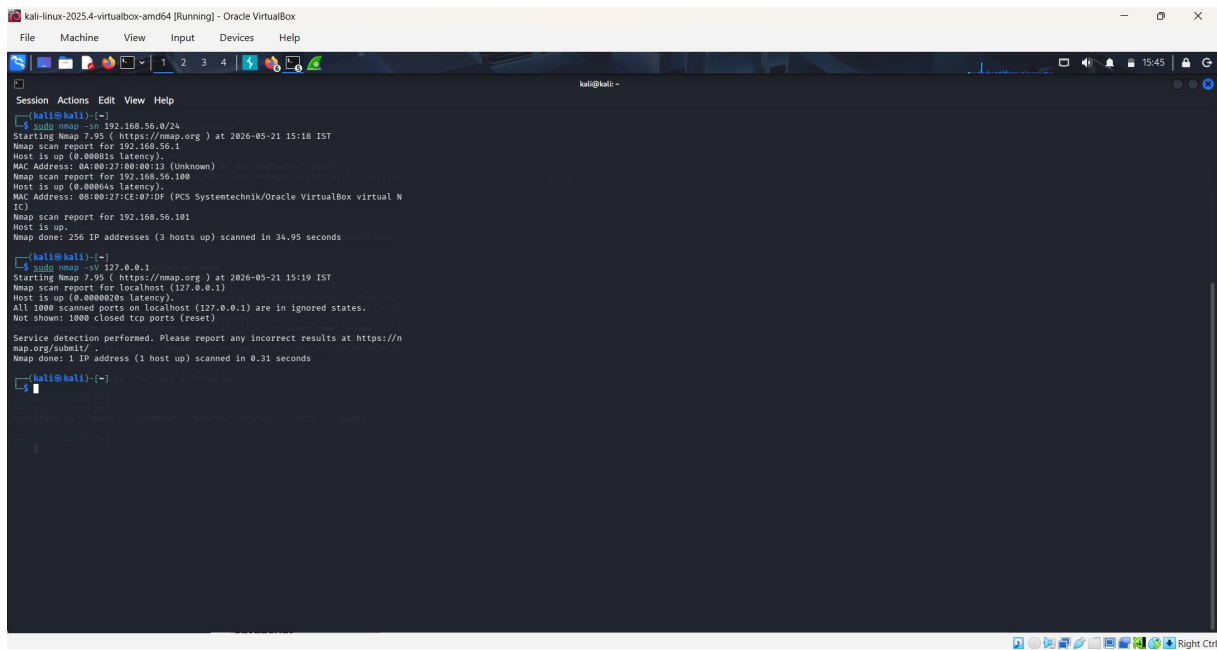
Figure 8: Nmap -sn ping sweep of 192.168.56.0/24 — 3 hosts discovered (gateway, host machine, Kali VM). Service scan on 127.0.0.1 also shown confirming loopback is isolated.

Scan 2 — Service Version Scan (localhost)

A service version scan was run against the loopback address (127.0.0.1) to confirm that no unintended services are exposed. All 1000 TCP ports were in a closed/reset state, confirming the loopback interface is properly isolated.

Command executed:

```
sudo nmap -sV 127.0.0.1
```



```
kali@kali:~$ sudo nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-21 15:18 IST
Nmap scan report for 127.0.0.1
Host is up (0.00001s latency).
MAC Address: 0A:00:27:00:00:13 (Unknown)
Nmap scan report for 127.0.0.1
Host is up (0.00004s latency).
MAC Address: 08:00:27:CE:07:DF (PCS Systemtechnik/Oracle VirtualBox virtual N
IC)
Nmap scan report for 127.0.0.1
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 34.95 seconds
kali@kali:~$ sudo nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-21 15:19 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000026s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Service detection performed. Please report any incorrect results at https://n
map.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.31 seconds
kali@kali:~$
```

Figure 9: Nmap -sV 127.0.0.1 — all 1000 ports in ignored states, confirming no unintended service exposure

8. Step 6 — Burp Suite Proxy Interception

Burp Suite Community Edition v2025.10.6 was launched and configured as an HTTP intercepting proxy. Firefox was configured to route traffic through Burp Suite on port 8080. This allows all HTTP/HTTPS requests and responses between the browser and web applications to be inspected, modified, and analysed in real time.

Burp Suite Configuration Summary:

Tool	Burp Suite Community Edition v2025.10.6
Listener Port	8080 (localhost)
Mode	Intercept ON — active proxy interception
Browser	Firefox configured with Burp proxy settings
Target	http://detectportal.firefox.com (connectivity test) & DVWA
Requests Captured	HTTP GET requests successfully intercepted

Screenshot 9: Burp Suite — Proxy Intercept Active

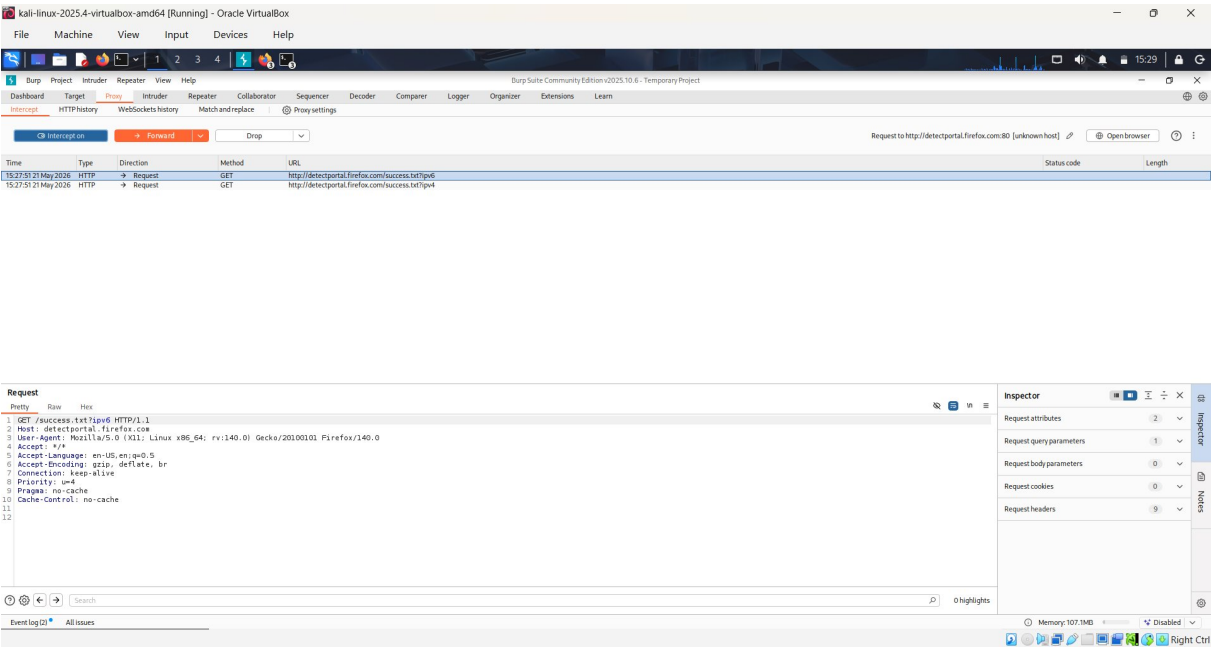


Figure 10: Burp Suite Community Edition — Intercept ON. HTTP GET request to detectportal.firefox.com captured showing full request headers

Screenshot 10: Burp Suite — HTTP History Log

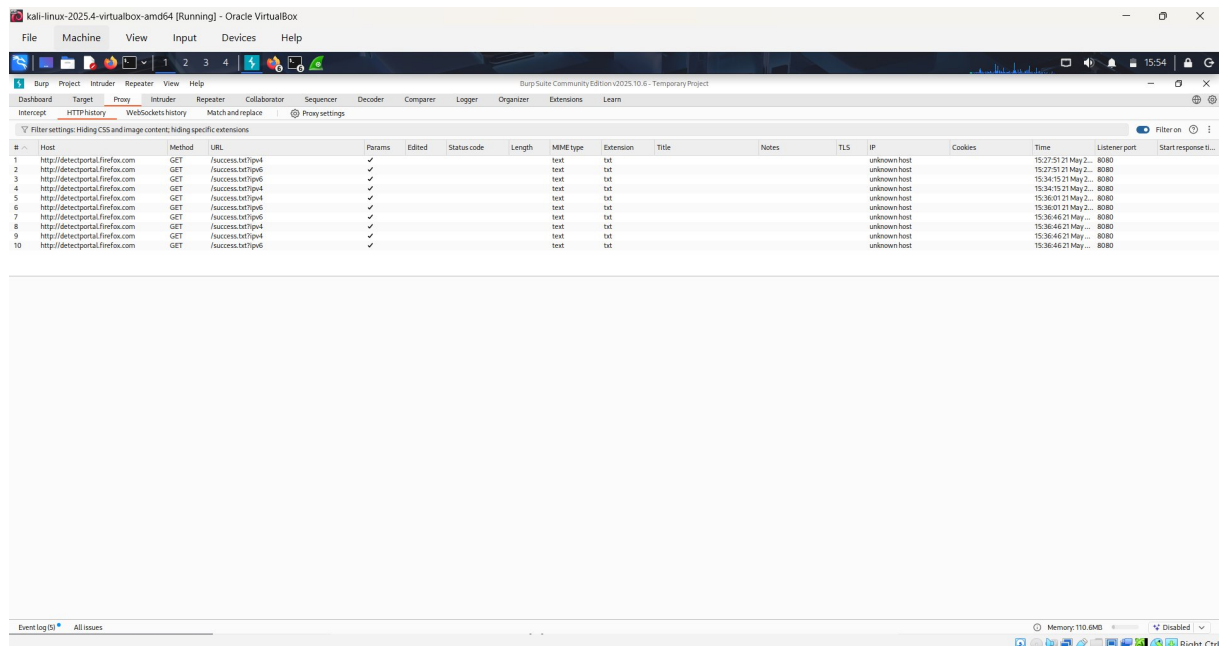


Figure 11: Burp Suite HTTP History — multiple intercepted requests to `detectportal.firefox.com` via port 8080, confirming proxy is fully operational

Screenshot 11: Burp Suite — Latest HTTP History (extended session)

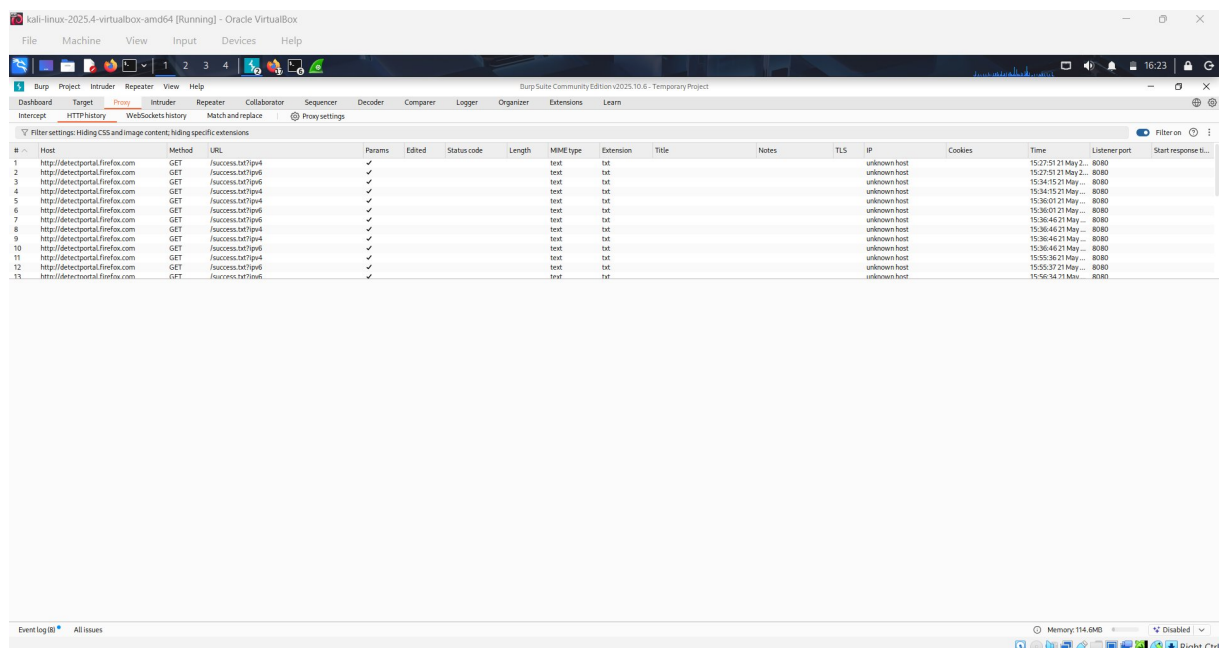


Figure 12: Extended Burp Suite HTTP history showing 13 captured entries across multiple sessions — proxy continuously intercepting traffic

Screenshot 12: Burp Suite — Intercept View (second session)

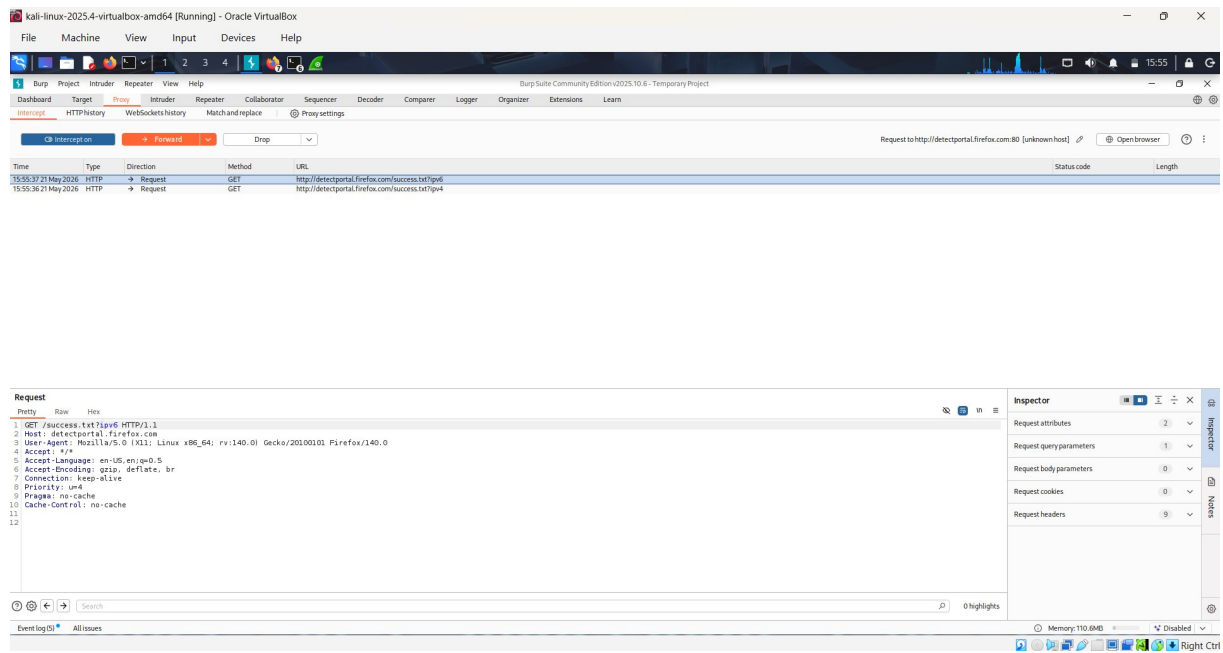


Figure 13: Burp Suite Proxy Intercept showing GET /success.txt?ipv6 HTTP/1.1 request with full Mozilla Firefox 140.0 User-Agent string

9. Step 7 — Wireshark Traffic Capture & Analysis

Wireshark was used to capture and analyse live network traffic. Two capture sessions were conducted: one on the eth0 (external) interface with an HTTP display filter, and one on the Loopback (lo) interface to capture HTTP traffic between the browser and the locally running DVWA application. Packet-level inspection confirmed the complete HTTP request-response cycle.

Capture Session 1 — eth0 Interface with HTTP Filter

Wireshark was started on the eth0 interface with an "http" display filter applied. The capture confirmed that HTTP traffic filtering is operational.

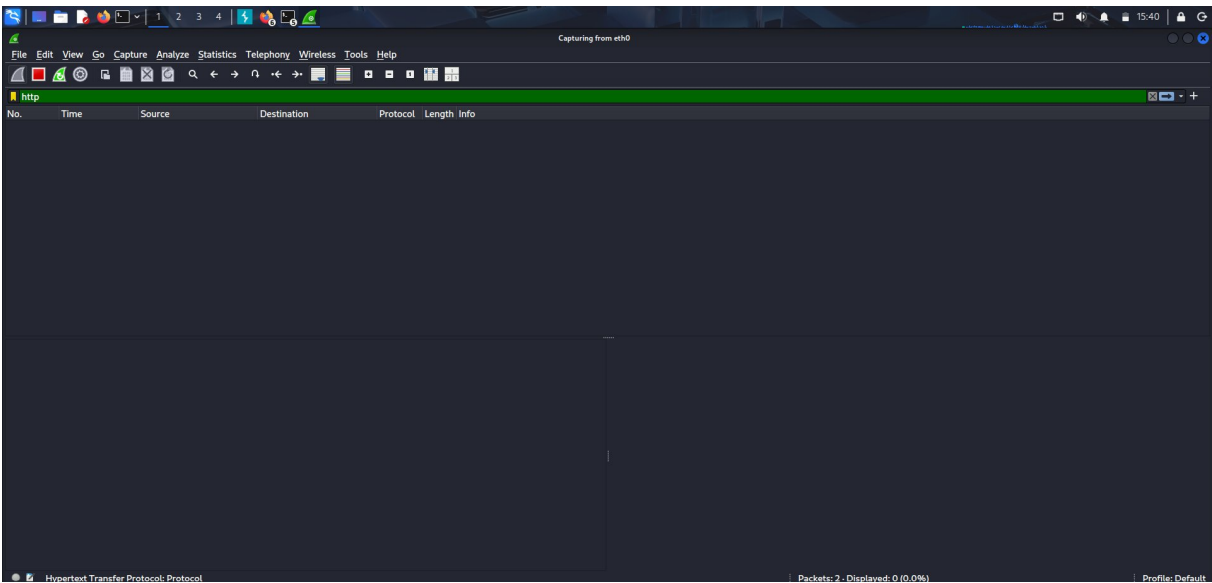


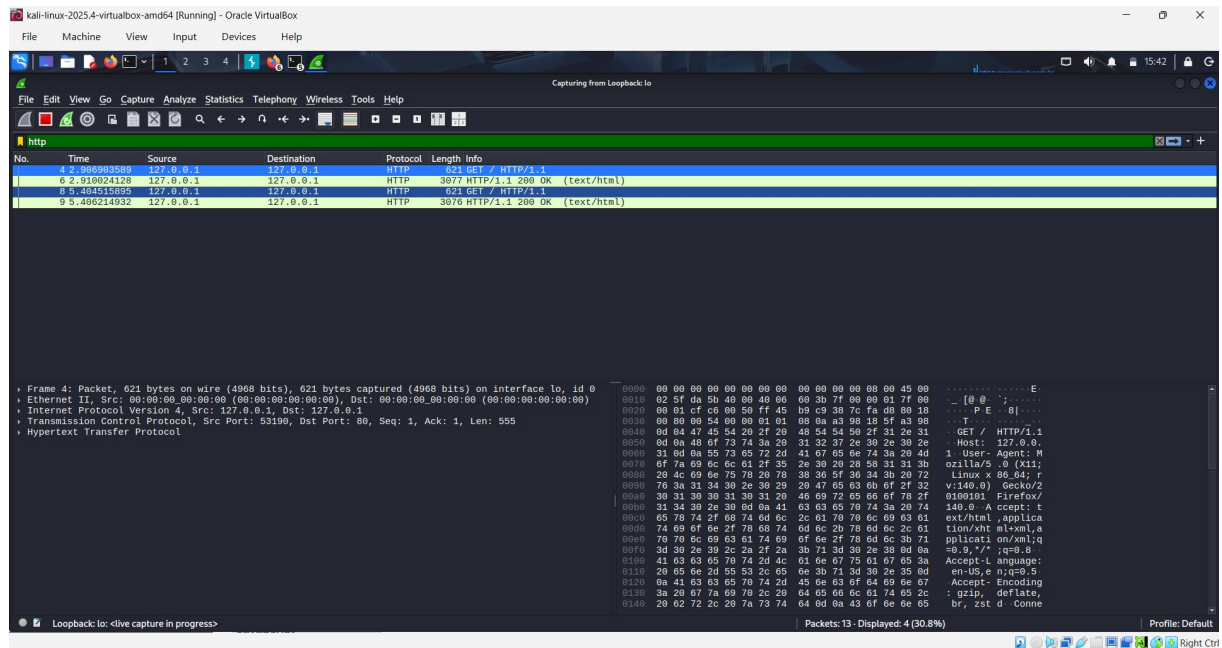
Figure 14: Wireshark capturing on eth0 interface with "http" display filter — capture running, waiting for HTTP packets

Capture Session 2 — Loopback Interface (DVWA Traffic)

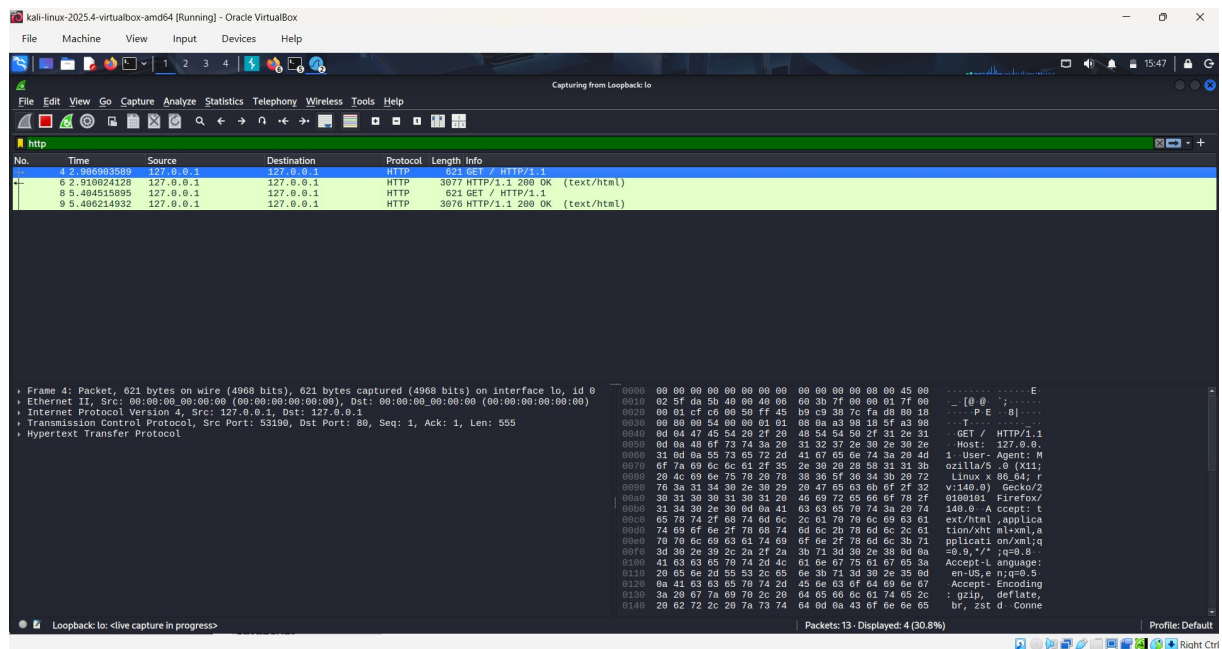
The capture interface was switched to the Loopback (lo) adapter to capture HTTP traffic between Firefox and the locally running DVWA. Four HTTP packets were captured and displayed (30.8% of 13 total packets). The capture shows a complete HTTP request/response exchange.

Capture Interface	Loopback: lo (127.0.0.1 ↔ 127.0.0.1)
Display Filter	http
Total Packets	13 captured, 4 displayed (30.8%)
Protocol	HTTP/1.1 over TCP
Source Port	53190 (browser) → Destination Port: 80 (Apache/DVWA)
Server Response	HTTP/1.1 200 OK — text/html (Apache/2.4.25)

Screenshot 13: Wireshark — HTTP Packets Captured (Frame 4 — GET Request)



Screenshot 14: Wireshark — Frame 6 (HTTP 200 OK Response)



Screenshot 15: Wireshark — Frame 8 (Second GET Request)

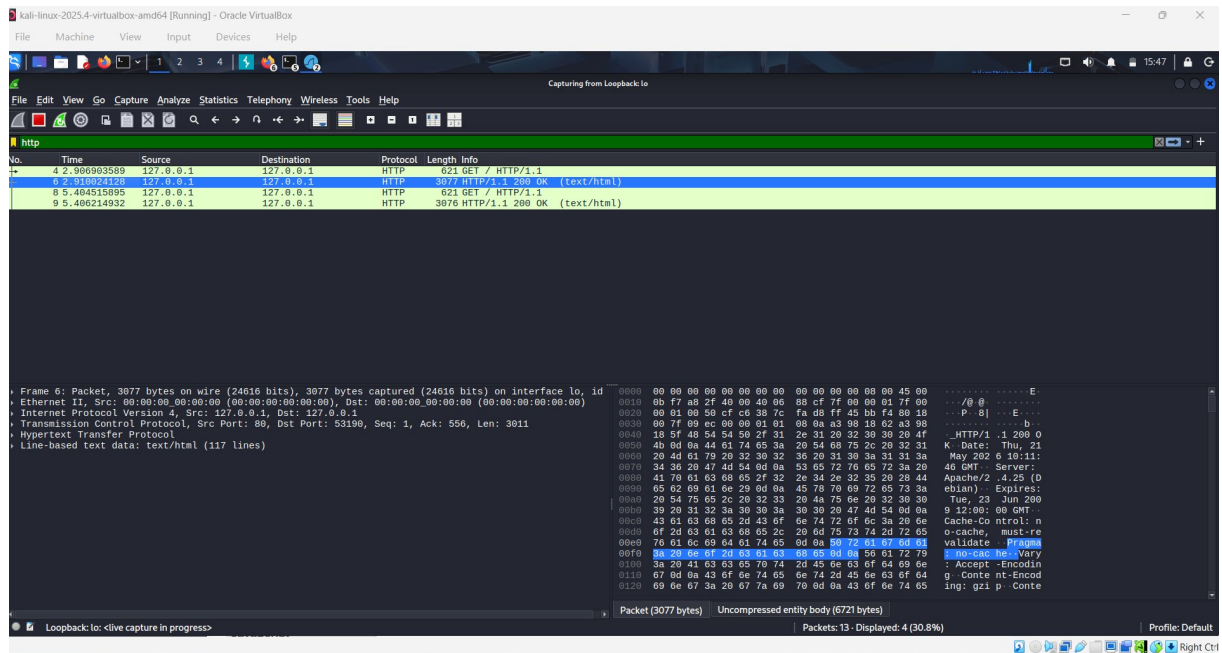


Figure 17: Frame 8: Second GET / HTTP/1.1 request (621 bytes) — browser reloading DVWA page. Packet shows Accept: text/html, application/xhtml+xml headers.

Screenshot 16: Wireshark — Frame 9 (Final HTTP Response)

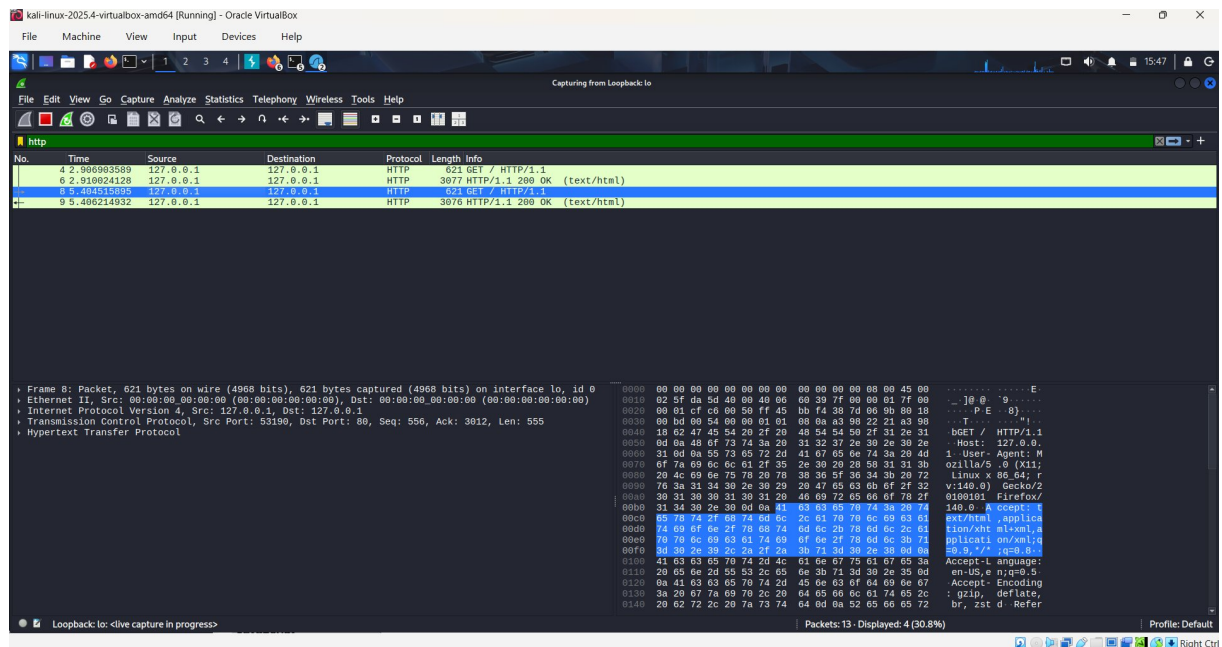


Figure 18: Frame 9: HTTP/1.1 200 OK response (3076 bytes, uncompressed 6721 bytes) — Apache server response with Vary: Accept-Encoding header. Complete request-response cycle confirmed.

Screenshot 17: Wireshark — Packet Detail View (HTTP Response Headers)

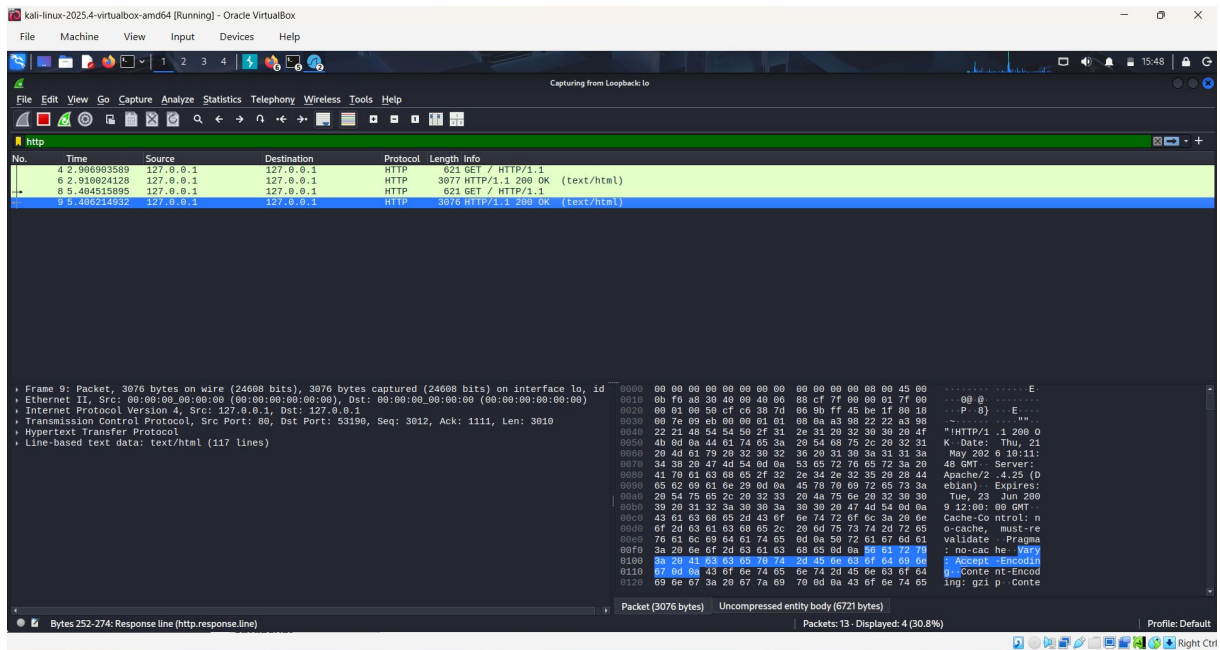


Figure 19: Detailed Wireshark packet view showing Response Line (`http.response.line`) — Server: Apache/2.4.25, Date: Thu 21 May 2026 10:11:48 GMT, highlighted in hex dump

10. Step 8 — DVWA Web Application Access & Validation

DVWA was accessed through the Firefox browser at <http://127.0.0.1>. The application was fully operational with all vulnerability modules available in the navigation menu. This confirms the complete and successful deployment of the target application within the lab environment.

DVWA Vulnerability Modules Available:

Module	Vulnerability Type
Brute Force	Login brute-force attacks
Command Injection	OS command injection via web forms
CSRF	Cross-Site Request Forgery
File Inclusion	Local/Remote file inclusion vulnerabilities
File Upload	Unrestricted file upload exploitation
Insecure CAPTCHA	CAPTCHA bypass techniques
SQL Injection	Database query manipulation
SQL Injection (Blind)	Time-based and boolean blind SQLi
Weak Session IDs	Session token prediction/hijacking
XSS (DOM)	Document Object Model-based XSS
XSS (Reflected)	Reflected Cross-Site Scripting
XSS (Stored)	Persistent Cross-Site Scripting
CSP Bypass	Content Security Policy bypass
JavaScript	Client-side JavaScript security

Screenshot 18: DVWA Home Page — All Modules Accessible

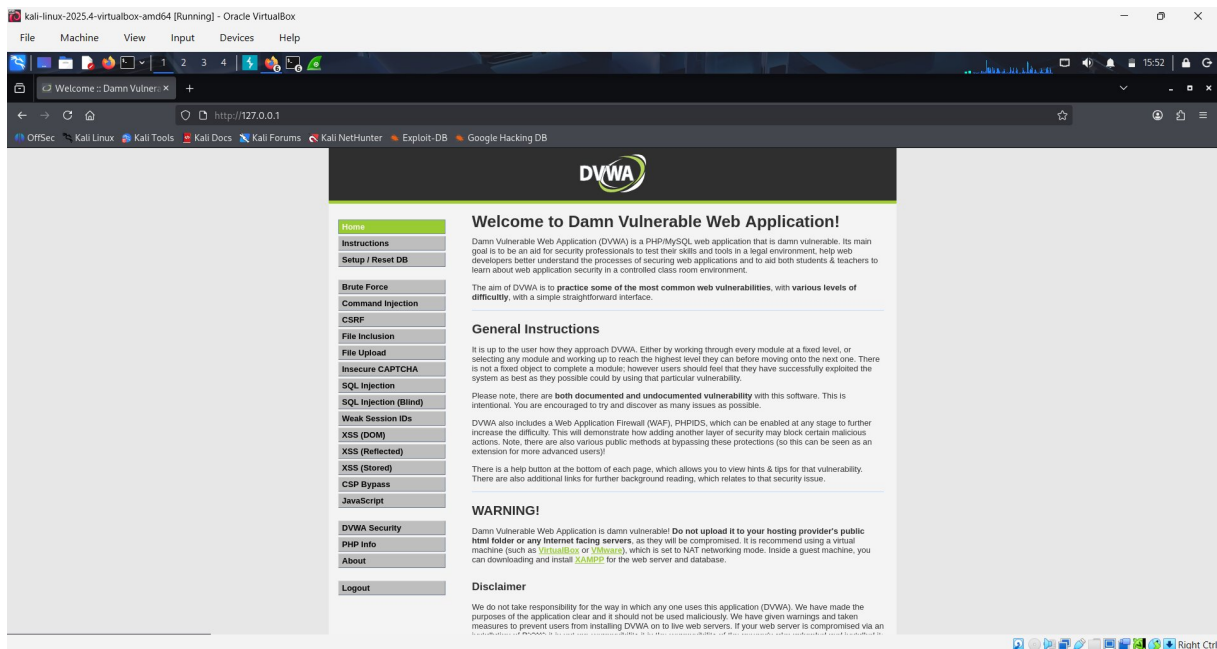


Figure 20: DVWA Welcome page at http://127.0.0.1 — all 14 vulnerability modules listed, Apache server confirmed, application fully operational

Screenshot 19: DVWA Login Page — Session Management Confirmed

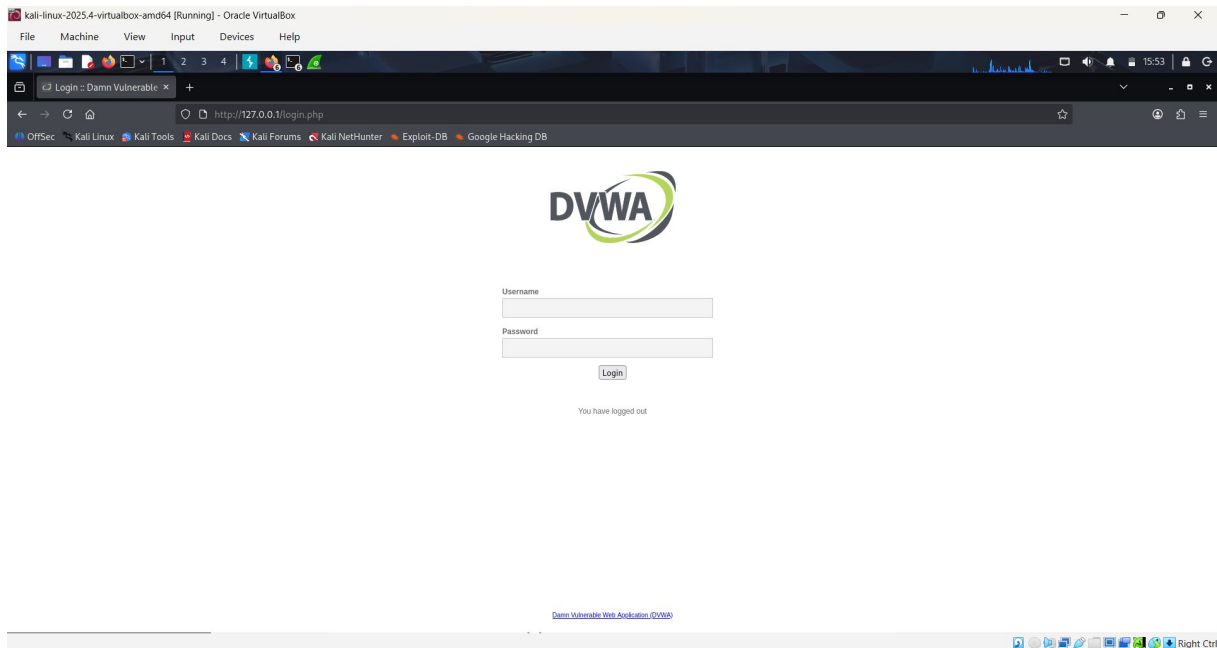


Figure 21: DVWA Login page at http://127.0.0.1/login.php — "You have logged out" message confirms session management is functional

11. Summary of Deliverables

All required deliverables specified in Task-2 have been completed and are evidenced by the screenshots included in this report.

VM List	Show virtual machines created	✓ Complete	Figures 1, 2 — VirtualBox Manager & Kali desktop
IP Configuration	Show network interfaces & IPs	✓ Complete	Figure 4 — ip addr showing all interfaces
Ping / Nmap Validation	Confirm attacker sees target	✓ Complete	Figures 8, 9 — Nmap scans (host discovery & service scan)
Burp Suite Screenshots	Intercept HTTP traffic	✓ Complete	Figures 10–13 — Burp intercept and history
Wireshark Screenshots	Capture and analyse packets	✓ Complete	Figures 14–19 — Full packet capture analysis
DVWA Deployment	Deploy vulnerable web application	✓ Complete	Figures 20, 21 — DVWA running at 127.0.0.1
PDF Report	Full lab build documentation	✓ Complete	This document

12. Conclusion

The personal cybersecurity lab has been successfully built and validated in accordance with all requirements set out in Maincrafts Technology Internship Task-2. The environment is fully isolated, safe, and ready for future security exercises including vulnerability assessment, web application penetration testing, and network traffic analysis.

Key Achievements:

- Successfully deployed Kali Linux 2025.4 as an attacker VM on VirtualBox with correct hardware allocation
- Configured dual-network adapters (NAT + Host-Only) providing both internet access and lab isolation
- Deployed DVWA as the intentionally vulnerable target application on Apache/PHP stack
- Validated network connectivity using Nmap — identified 3 hosts on the 192.168.56.0/24 subnet
- Confirmed HTTP traffic interception using Burp Suite Community Edition proxy
- Captured and analysed HTTP packets at frame level using Wireshark on the loopback interface
- Verified complete HTTP request-response cycle between browser and DVWA application
- All tools (Nmap, Burp Suite, Wireshark) validated and operational for future use

This lab environment provides a solid, isolated foundation for all subsequent cybersecurity tasks and practical assessments during this internship programme.

Submitted By:

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